Volume

Name _____ **Date** _____ No calculator Version 1

KEY WORDS volume · cube · rectangular prism · length · width · height · cubic units

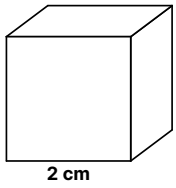
A Cubes ●○○

WORK IN YOUR BOOK

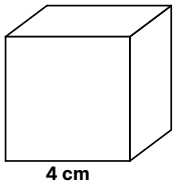
Remember: Volume is the amount of space inside a solid, measured in cubic units such as cm^3. Every edge of a cube is the same length, so $V = l \times l \times l = l^3$.

1. Find the volume of each cube.

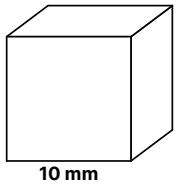
a)



b)



c)



2. Find the volume of:

a) A cube with edges of 3 m **b)** A cube with edges of 6 cm **c)** A cube with edges of 1 m

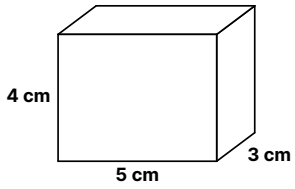
B Rectangular prisms ●●○

WORK IN YOUR BOOK

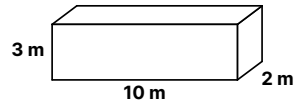
Remember: Volume of a rectangular prism = length × width × height.

3. Find the volume of each rectangular prism.

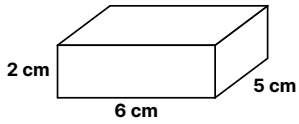
a)



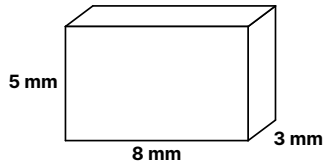
b)



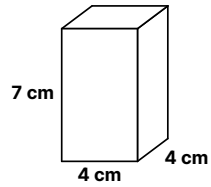
c)



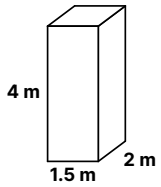
d)



e)



f)



4. Find the volume of:

a) A rectangular prism 12 cm by 5 cm by 2 cm **b)** A rectangular prism 20 cm by 10 cm by 5 cm

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C Missing dimensions ●●● — WORK IN YOUR BOOK

> **Remember:** Work backwards. If volume = length × width × height, then height = volume ÷ (length × width).

5. Find the missing length.

a) A rectangular prism has a volume of 120 cm³. It is 6 cm long and 5 cm wide. Find its height.

b) A cube has a volume of 27 cm³. Find the length of its edges.

c) A rectangular prism has a volume of 84 m³. It is 7 m long and 4 m wide. Find its height.

d) A cube has a volume of 64 cm³. Find the length of its edges.

e) A rectangular prism has a volume of 200 cm³. It is 10 cm long and 4 cm wide. Find its height.

D Units for length, area and volume ●●○ — ANSWER ON THE SHEET

> **Remember:** A length is measured in units (cm). An area is measured in square units (cm²). A volume is measured in cubic units (cm³).

6. Which unit would you use for each of these: cm, cm² or cm³?

a) The height of a cup _____

b) The amount of space inside a box _____

c) The size of a page in a book _____

d) The distance around a plate _____

e) The amount of wrapping paper that covers a present _____

f) The amount of sand in a sandpit _____

E Problems ●●● — WORK IN YOUR BOOK

7. A fish tank is 50 cm long, 30 cm wide and 20 cm deep. Find its volume.

8. A box is 4 cm by 3 cm by 2 cm. Find the total volume of 5 of these boxes.

★ Extension — FOR EARLY FINISHERS

9. A cube has edges of 3 cm. Every edge is doubled. How many times larger is the volume of the new cube?

10. How many cubes with edges of 2 cm fit exactly into a box that is 6 cm by 4 cm by 2 cm?

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Volume

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS volume · cube · rectangular prism · length · width · height · cubic units

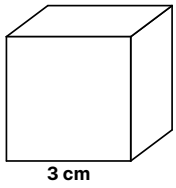
A Cubes ●○○

WORK IN YOUR BOOK

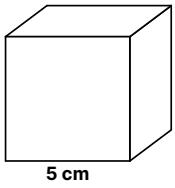
Remember: Volume is the amount of space inside a solid, measured in cubic units such as cm^3. Every edge of a cube is the same length, so $V = l \times l \times l = l^3$.

1. Find the volume of each cube.

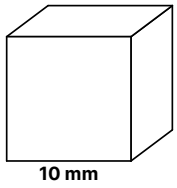
a)



b)



c)



2. Find the volume of:

a) A cube with edges of 2 m **b)** A cube with edges of 7 cm **c)** A cube with edges of 1 m

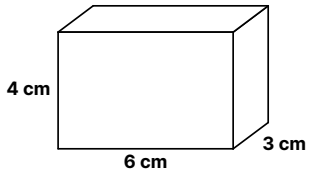
B Rectangular prisms ●●○

WORK IN YOUR BOOK

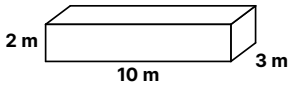
Remember: Volume of a rectangular prism = length × width × height.

3. Find the volume of each rectangular prism.

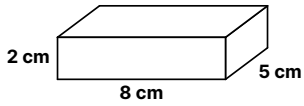
a)



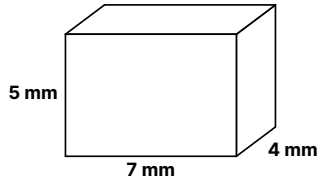
b)



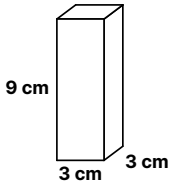
c)



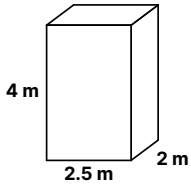
d)



e)



f)



4. Find the volume of:

a) A rectangular prism 15 cm by 4 cm by 2 cm **b)** A rectangular prism 30 cm by 10 cm by 5 cm

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C Missing dimensions ●●● — WORK IN YOUR BOOK

> **Remember:** Work backwards. If volume = length × width × height, then height = volume ÷ (length × width).

5. Find the missing length.

a) A rectangular prism has a volume of 160 cm³. It is 8 cm long and 5 cm wide. Find its height.

b) A cube has a volume of 8 cm³. Find the length of its edges.

c) A rectangular prism has a volume of 72 m³. It is 6 m long and 4 m wide. Find its height.

d) A cube has a volume of 125 cm³. Find the length of its edges.

e) A rectangular prism has a volume of 300 cm³. It is 10 cm long and 6 cm wide. Find its height.

D Units for length, area and volume ●●○ — ANSWER ON THE SHEET

> **Remember:** A length is measured in units (cm). An area is measured in square units (cm²). A volume is measured in cubic units (cm³).

6. Which unit would you use for each of these: cm, cm² or cm³?

a) The width of a door _____ **b)** The amount of space inside a drawer _____

c) The size of a phone screen _____ **d)** The distance around a photo _____

e) The amount of paint that covers a wall _____ **f)** The amount of water in a fish tank _____

E Problems ●●● — WORK IN YOUR BOOK

7. A fish tank is 60 cm long, 25 cm wide and 20 cm deep. Find its volume.

8. A box is 5 cm by 2 cm by 2 cm. Find the total volume of 6 of these boxes.

★ Extension — FOR EARLY FINISHERS

9. A cube has edges of 2 cm. Every edge is doubled. How many times larger is the volume of the new cube?

10. How many cubes with edges of 2 cm fit exactly into a box that is 8 cm by 4 cm by 2 cm?

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